DEEPAK R V

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Professional Summary

Computer Science graduate specializing in Artificial Intelligence and Machine Learning with hands-on experience in computer vision, AI systems, XR integration, software development, and experiment-driven R&D. Strong problem-solving skills with experience in developing, validating, debugging, and optimizing real-world technical prototypes.

Education

S.A. Engineering College, Chennai 2022 – 2026

Bachelor of Engineering in Computer Science and Engineering (AI/ML Specialization) **CGPA: 7.99 / 10**

- **Activities & Societies:** IEEE Student Member; coordinated a department-level hackathon; participated in technical events and project exhibitions.
- **Coursework:** Data Structures, Algorithms, Database Management Systems, Machine Learning, Artificial Intelligence, Computer Vision.

Experience

Computer Vision / AI R&D Intern – BOA XR Sep 2026 – Oct 2026

iTNT Hub, Anna University, Guindy | Confidential Real-Time XR Prototype

- Conducted hands-on Computer Vision and AI R&D by analyzing an inherited codebase, validating baseline behavior, and using controlled experiments to identify technical limitations.
- Developed and evaluated vision-based physical measurement workflows involving camera calibration, object localization, segmentation, geometric analysis, and independent ground-truth validation.
- Integrated computer-vision processing with Unity-based XR workflows and real hardware, validating perception, spatial feedback, and runtime behavior under device conditions.
- Performed systematic failure analysis across detection, segmentation, geometry, recognition, tracking, and runtime stages; applied regression testing and documented fixes, limitations, and findings.
- Profiled live pipeline behavior, identified processing bottlenecks, and implemented runtime optimizations while verifying that changes preserved expected system behavior.

Frontend Developer – FORCRUX May 2026 – Jul 2026

Part-time | Four-Member Team Startup Initiative

- Worked as the frontend developer in a four-member team under FORCRUX, developing the Kaatchi Media client website based on business, design, and content requirements.
- Developed responsive interfaces, photography galleries, portfolio filtering, animations, reusable UI components, and interactive navigation using modern frontend practices.
- Contributed to testing, debugging, performance optimization, Git/GitHub collaboration, and production deployment of the client website.

Intern – Embedded Systems Aug 2024 – Sep 2024

MSME, Guindy, Chennai

- Developed automation projects using Arduino boards and IoT components for real-time monitoring and control.
- Programmed sensors and output modules and worked on microcontroller integration, circuit debugging, and prototype testing.
- Gained practical exposure to embedded hardware workflows, system troubleshooting, and real-time control applications.

Projects

SightMate – AI Navigation Assistant for the Visually Impaired

GitHub: github.com/DEEPAKRV07/SightMate-AI-Assistant

- Developed a Flutter-based Android application combining on-device computer vision, navigation assistance, and accessible voice interaction for visually impaired users.
- Integrated YOLOv8 TensorFlow Lite with Fast-SCNN semantic segmentation and designed a fusion pipeline to identify obstacles, walkable paths, and directional guidance.
- Implemented Google ML Kit OCR, voice commands, Braille input, translation, and Text-to-Speech within a modular offline AI architecture.
- **Tech Stack:** Flutter, Dart, TensorFlow Lite, YOLOv8, Fast-SCNN, Google ML Kit, OCR, Speech-to-Text, Text-to-Speech

Football Match Video Analysis using YOLOv8 & ByteTrack
GitHub: github.com/DEEPAKRV07/Football-Analysis-System

- Developed an end-to-end computer vision pipeline using YOLOv8 and ByteTrack to detect and track players, referees, goalkeepers, and the ball across match videos.
- Applied camera-motion estimation, view transformation, jersey-color clustering, and ball trajectory handling to improve tracking and team-level analysis.
- Calculated player speed, distance covered, and team ball possession and generated annotated match videos with tracking IDs and analytical overlays.
- **Tech Stack:** Python, OpenCV, YOLOv8, ByteTrack, Optical Flow, Scikit-learn, NumPy, Pandas

Google Maps Lead Generation & Intelligence Platform
GitHub: github.com/DEEPAKRV07/Google-Maps-Lead-Generator

- Developed an automated lead generation platform using Playwright to collect Google Maps business data and enrich leads with website contact information.
- Implemented SQLite-based persistence, resumable checkpoints, duplicate detection, multithreaded website crawling, strict email validation, and automated Excel/CRM exports.
- Built AI-assisted website analysis and lead prioritization to identify digital gaps and generate explainable prospect scores and outreach recommendations.
- **Tech Stack:** Python, Playwright, Chromium, SQLite, Multithreading, Web Scraping, AI/LLMs, Excel, CRM Integration

Punch Hero – Legacy Unity Android Restoration & ARM64 Porting Study
GitHub Repository (Private)

- Conducting a personal, fan-driven compatibility study to analyze a legacy Unity 3.5.7f6 Android game and migrate its project to Unity 2022.3.62f3.
- Built and installed an ARM64/IL2CPP APK while preserving the original package identity and existing saved career data.
- Fixed Training Center prefab loading linked to duplicate serialized IDs and corrected inter-round camera timing; validated selected gameplay flows on a physical Android device.
- Restoration remains in progress: Mega Punch activation, character presentation, complete career progression, and broader gameplay validation are unresolved. No public APK is being claimed.
- *Tech Stack:* Unity, C#, Android, IL2CPP, ARM64, APK Analysis, ADB, PowerShell, Git, GitHub Actions

Technical Skills & Exposure

Programming: Python, SQL, C++, C, Dart, JavaScript, HTML5, CSS3

AI/ML: Machine Learning, Deep Learning, Computer Vision, Object Detection, Semantic Segmentation, Multi-Object Tracking, YOLOv8, Fast-SCNN, TensorFlow, TensorFlow Lite, OCR, K-Means Clustering

Computer Vision & Geometry: Camera Calibration, Image Analysis, Geometric Measurement, Stereo Vision, 3D Geometry, Feature Extraction, Ground-Truth Validation, Error Analysis

XR & Software: Unity, XR/MR Integration, Spatial Feedback, Real-Time Hardware Validation, Flutter, React, Next.js, TypeScript, Django, OpenCV, Google ML Kit, REST APIs

Generative AI & Exposure: LLMs, RAG, LangChain, LangGraph, Prompt Engineering, Embeddings, Vector Databases, AI Agents

Engineering & Tools: Git, GitHub, Regression Testing, Performance Profiling, Root-Cause Analysis, Experiment Design, Technical Documentation, Android Studio, Playwright, Chromium, Three.js, GSAP, SQL Server, SQLite3, Power BI